

2.9 PROTOCOL vs. HARDWARE: IS IT REALLY A SCANNER PROBLEM?

A SYSTEMATIC APPROACH TO IMAGE QUALITY & PERFORMANCE ISSUES

Most image-quality and performance problems are not hardware failures.

Follow this step-by-step path: start with the easiest and most common causes first.

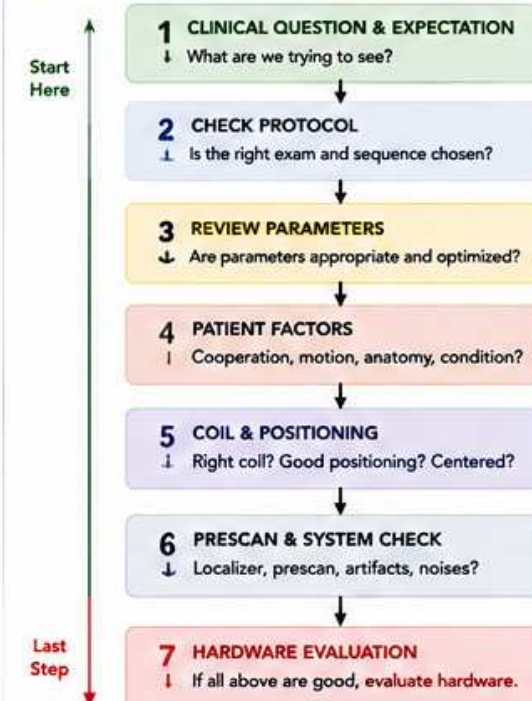


KEY CONCEPT

Check the simple things first.
Optimize the controllable parameters.
Only after that, suspect hardware.

↑ Increase / Improve
↓ Decrease / Reduce
— No / Minimal Effect

1 THE TROUBLESHOOTING PATH



2 COMMON PROBLEMS & LIKELY CAUSES

Problem / Symptom	Most Likely Causes (in order)	Check First
Blurry / Low Detail	- Large slice thickness - Low matrix - High BW - Motion / patient	- Parameters - Patient
Noisy Image	- Low NEX / averages - High parallel imaging (R) - High BW - Small coil / off-center	- Parameters - Coil - Positioning
Aliasing / Wrap	- FOV too small - Phase oversampling low - Wrong phase direction - High R factor	- Parameters - Protocol
Ghosting	- Motion - Flow - Eddy currents - Poor gating	- Patient - Parameters
Saturation / Poor Contrast	- Wrong sequence - Wrong TR / TE / TI - Wrong flip angle - B1 inhomogeneity	- Protocol - Parameters
Streaks / Lines	- Metal / susceptibility - Gradient / RF interference - Defective coil element - A/D or cable issue	- Coil - Prescan - Hardware

3 QUICK CHECKS (DO THESE FIRST)

Protocol	✓ Right exam / sequence for the clinical question? ✓ Contrast and tissue weighting appropriate? ✓ 2D vs 3D – was the correct choice made? ✓ Field strength appropriate?
Parameters	✓ Slice thickness, matrix, FOV appropriate? ✓ NEX / averages adequate? ✓ BW not unnecessarily high? ✓ ETL / TSE factor appropriate? ✓ Parallel imaging factor reasonable?
Patient	✓ Can the patient lie still? ✓ Breath-hold / gating possible? ✓ Claustrophobia / pain managed? ✓ Implants / safety considerations?
Coil & Positioning	✓ Correct coil for the exam? ✓ Coil connected and elements active? ✓ Patient centered (L/R, A/P, S/I)? ✓ Anatomy in the FOV center?
Prescan & System	✓ Localizer complete and accurate? ✓ Prescan lines look clean? ✓ No obvious artifacts or noise? ✓ Gradient / RF noises normal? ✓ System messages / warnings?

4 WHEN TO SUSPECT HARDWARE

Possible Hardware Issue	What You May See
Defective Coil Element	- Localized signal void / drop-out - Streaks that move with position - Prescan element failure
Magnet / Shim	- Global distortion - Poor shimming (line broadening) - Inhomogeneous fat sat
RF System	- No transmit / receive - Intermittent dropouts - Heating / power errors
Gradients	- Streaks, geometric distortion - Excessive gradient noise - Poor fat/water separation
System / Electronics	- Frequent errors / resets - Timing issues - A/D or recon problems



Confirm that all **earlier steps (1–6)** have been checked and documented **before calling it a hardware issue.**

5 EXAMPLES: GOOD vs. NOT OPTIMIZED

Blurry (Low Resolution)		Aliasing (Wrap)		Noisy Image	
Not Optimized	Optimized	Not Optimized	Optimized	Not Optimized	Optimized
Thick slice, low matrix, high BW, low NEX	Thin slice, higher matrix, appropriate BW, adequate NEX	FOV too small, phase wrap	Larger FOV, no wrap	Low NEX, high R factor	Higher NEX, lower R factor

6 DOCUMENT WHAT YOU DID

	Good documentation saves time and helps others.
✓	Problem / symptom
✓	What you checked
✓	Parameter changes made
✓	Results
✓	Final solution or next step
	<i>If it isn't documented, it didn't happen.</i>

7 PRACTICAL TIPS

- Change one variable at a time.
- Small changes > big changes.
- Compare side-by-side when possible.
- Use prior images / protocols as reference.
- Talk to the patient — they matter.
- Think simple → complex → hardware.

8 ENGINEER CHECKLIST

- ✓ Have protocol and parameters been verified?
- ✓ Is the issue repeatable and consistent?
- ✓ Have patient, coil, and positioning been ruled out?
- ✓ Do prescan and system performance look normal?
- ✓ Are QC and preventive maintenance up to date?
- ✓ Only after all above → investigate hardware.



KEY TAKEAWAY

Follow the path. Be systematic. Don't guess.
Most problems are solved long before hardware.